

Target dossier

VEGFA: Vascular endothelial growth factor A

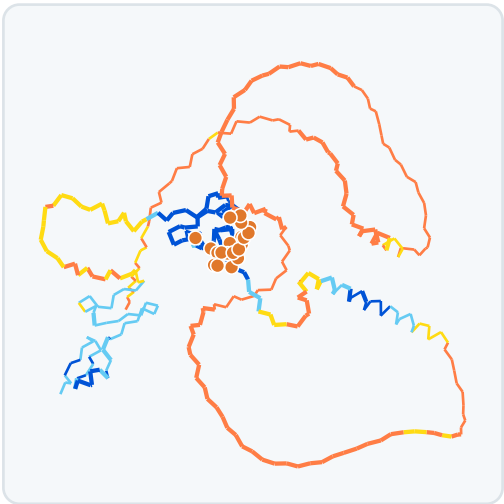
UniProt P15692 · 395 residues · ChEMBL CHEMBL1783 · linked findings: Referable diabetic retinopathy, Macular edema signal

For patient this screening: priority 98. Referable diabetic retinopathy (validated retinal model); Anti-VEGF intravitreal therapy: specialist-administered; 6 approved drugs; drug-contact site pLDDT 95 (PDB 1CZ8).

Key insights

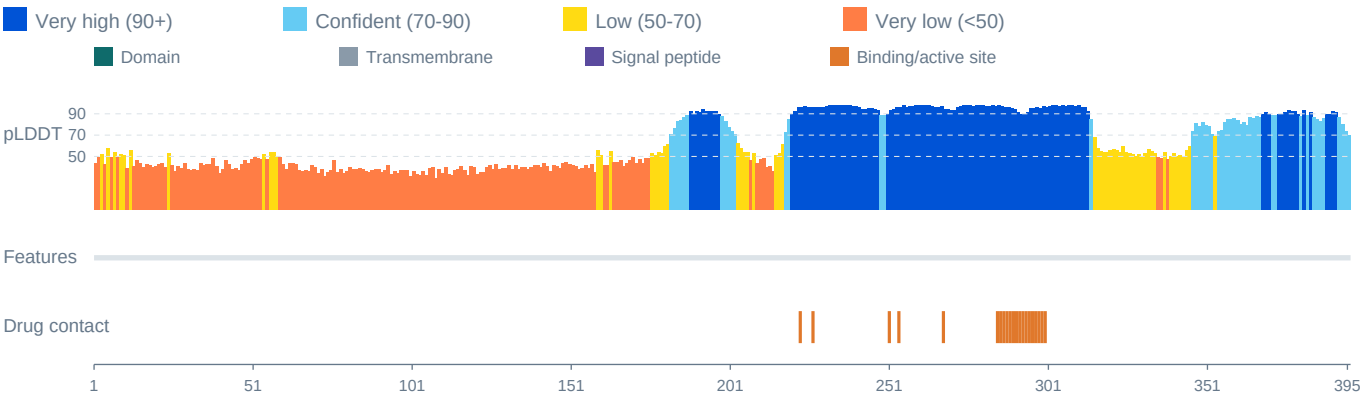
- **Druggable site mapped.** 21 residues lie within 4.5 Å of the antibody or peptide chains H, L, X, Y in PDB 1CZ8; AlphaFold confidence there is 95 (high: the AlphaFold model reproduces the site well enough for structure-based screening and docking).
- **Flexible regions** (pLDDT below 50) at residues 13-23, 25-53, 59-158, 164-175: likely disordered or unstructured; exclude them from pocket searches and expect them to be absent in crystal structures.
- **Validated and crowded mechanism.** 6 approved drugs (antibody 4, protein 3, small molecule 1, unknown 1); a new entrant needs differentiation in selectivity, dosing interval, delivery or a defined patient subgroup.
- **Patient demand signal.** 4 of 21 screened patients (19%) carry a finding linked to this target: a recruitment and market signal detected non-invasively.
- **Clinical activity.** 8 recruiting trial(s) near the clinic for the linked findings.

Structure



Mean pLDDT	63.9
Confident residues (pLDDT 70+)	42%
Disordered regions	13-23, 25-53, 59-158, 164-175
Drug-contact residues	21 (PDB 1CZ8)
Contact-site pLDDT	94.6

AlphaFold model projected on its two principal axes, coloured by confidence; orange dots mark residues touching the drug in the experimental complex.



Drug-contact residues (UniProt numbering): F223, Y227, Y251, K254, N268, Q285, I286, M287, R288, I289, K290, P291, H292, Q293, G294, Q295, H296, I297, G298, E299, M300

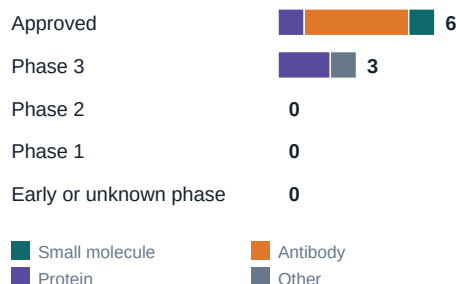
Function (UniProt)

Participates in the induction of key genes involved in the response to hypoxia and in the induction of angiogenesis such as HIF1A.

Why the eye points here

Hypoxic retina releases VEGF-A, which makes vessels leak (macular edema) and drives new fragile vessels (proliferative DR). Referable DR and the macular edema signal are the retinal phenotype of VEGF-driven disease.

Drug landscape



Drug	Stage	Action	Modality	ChEMBL
afibercept	Approved	inhibitor	protein	CHEMBL1742982
bevacizumab	Approved	inhibitor	antibody	CHEMBL1201583
brolicizumab	Approved	inhibitor	antibody	CHEMBL3707357
faricimab	Approved	inhibitor	antibody	CHEMBL4297750
pegaptanib	Approved	antagonist	small molecule	CHEMBL1201421
ranibizumab	Approved	inhibitor	antibody	CHEMBL1201825
abicipar pegol	Phase 3	inhibitor	protein	CHEMBL4297684
conbercept	Phase 3	inhibitor	protein	CHEMBL2108313
muparfostat	Phase 3	inhibitor	unknown	CHEMBL4297163

In the Insight Rx therapy guide

- **Anti-VEGF intravitreal therapy** (Specialist-administered): First-line for centre-involved diabetic macular edema with vision loss and an option for proliferative DR. Given by a retina specialist, so the referral is the action. AAO Diabetic Retinopathy Preferred Practice Pattern; DRCR Retina Network Protocols T and S

Recruiting clinical trials

ClinicalTrials.gov, searched by finding near the clinic; no patient data is sent. The site confirms full eligibility.

Finding	Trial	Phase / sponsor	Sites
Referable diabetic retinopathy	A Study to Investigate the Safety, Tolerability, and Pharmacokinetics of RO7663498 Following Intravitreal Administration in Participants With Diabetic Retinopathy NCT07588100 Global Research Management, Glendale, California	Phase 1 · Hoffmann-La Roche	3 site(s)
Referable diabetic retinopathy	Implementing AI-based Glaucoma Screening Within Federally Qualified Health Centers NCT06629649 Department of Ophthalmology and Visual Sciences, University of Alabama at Birmingham, Birmingham, Alabama	NA · University of Alabama at Birmingham	1 site(s)
Referable diabetic retinopathy	Longitudinal Natural History Study of Retinal Function in Eyes of Patients With Diabetes NCT07270133 Southeast Retina Center, P.C., Augusta, Georgia	N/A · Jaeb Center for Health Research	6 site(s)
Referable diabetic retinopathy	Biomarker of Diabetic Retinopathy NCT05079399 Spring Mill Clinic, Carmel, Indiana	N/A · Indiana University	3 site(s)
Macular edema signal	A Study in People With Diabetic Macular Edema to Test How Well Different Doses of BI 3812465 Are Tolerated NCT07553429 Associated Retina Consultants, Ltd., Phoenix, Arizona	Phase 1, Phase 2 · Boehringer Ingelheim	29 site(s)

Finding	Trial	Phase / sponsor	Sites
Macular edema signal	A Study to Evaluate the Safety, Tolerability, Pharmacodynamics, and Pharmacokinetics of RO7823653 in Participants With Diabetic Macular Edema (DME) NCT07425522 Associated Retina Consultants - Peoria - DocTrials - PPDS, Peoria, Arizona	Phase 1 · Genentech, Inc.	12 site(s)
Macular edema signal	A Clinical Trial of MK-8748 Compared to Aflibercept in Participants With Diabetic Macular Edema (MK-8748-004) NCT07705555 Global Research Management (Site 0066), Glendale, California	Phase 3 · Merck Sharp & Dohme LLC	8 site(s)
Macular edema signal	Surabgene Lomparvovec Administered in the Suprachoroidal Space in Adult Participants With Diabetic Retinopathy Without Center-Involved Diabetic Macular Edema NCT07592273 Retinal Consultants Of Arizona - Phoenix - North /ID# 283407, Phoenix, Arizona	Phase 2, Phase 3 · AbbVie	34 site(s)

Sources

UniProt; AlphaFold Protein Structure Database (DeepMind/EMBL-EBI, CC-BY 4.0); RCSB PDB; ChEMBL mechanisms (EMBL-EBI); ClinicalTrials.gov. Contacts computed at 4.5 Å between heavy atoms of the drug and the target, mapped to UniProt numbering by sequence alignment.